

MATHEMATICS DEPARTMENT

GENERAL MATHEMATICS

STRENGTHENED SENIOR HIGH SCHOOL CURRICULUM



OPENING PRAYER



**INSERT
ATTENDANCE LINK
(IF AVAILABLE)**

CLASSROOM RULES

S Show respect

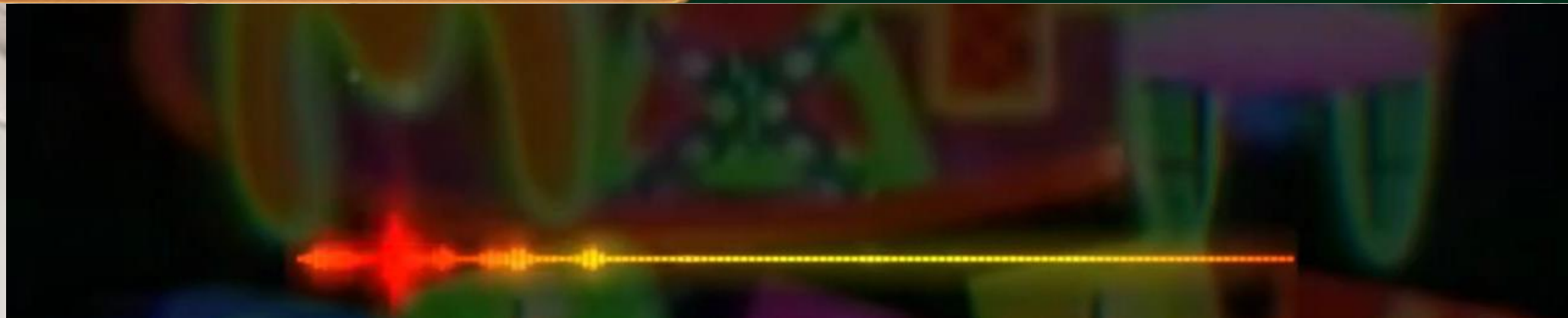
T Time Management and Attendance

E Engage in Learning

M Make safe and responsible choices

CLASSROOM ICE BREAKER: MATH-TINIK THEME SONG





MATH THINK THEME SONG





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RECAP

STRENGTHENED SENIOR HIGH SCHOOL CURRICULUM

**RECAP**

It is a single function defined by multiple sub-functions (or sub-rules), where each sub-function applies to a specific interval or domain.



RECAP

- Given the function:

$$f(x) = \begin{cases} 2x + 1 & \text{if } x < 0 \\ x^2 + 3 & \text{if } x \geq 0 \end{cases}$$

Find: $f(2)$, $f(-5)$, $f(-1)$

PROBLEM SOLVING INVOLVING PIECEWISE FUNCTION

**PRESENTED BY:
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Teacher V**

LESSON OBJECTIVES

- 🎯 Analyze a given real-world scenario (e.g., fare rates, purchasing, and income tax) to identify the distinct sub-domains and their corresponding mathematical rules.

LESSON OBJECTIVES

 **Formulate and evaluate
piecewise equations
accurately to calculate
specific values.**



LESSON OBJECTIVES

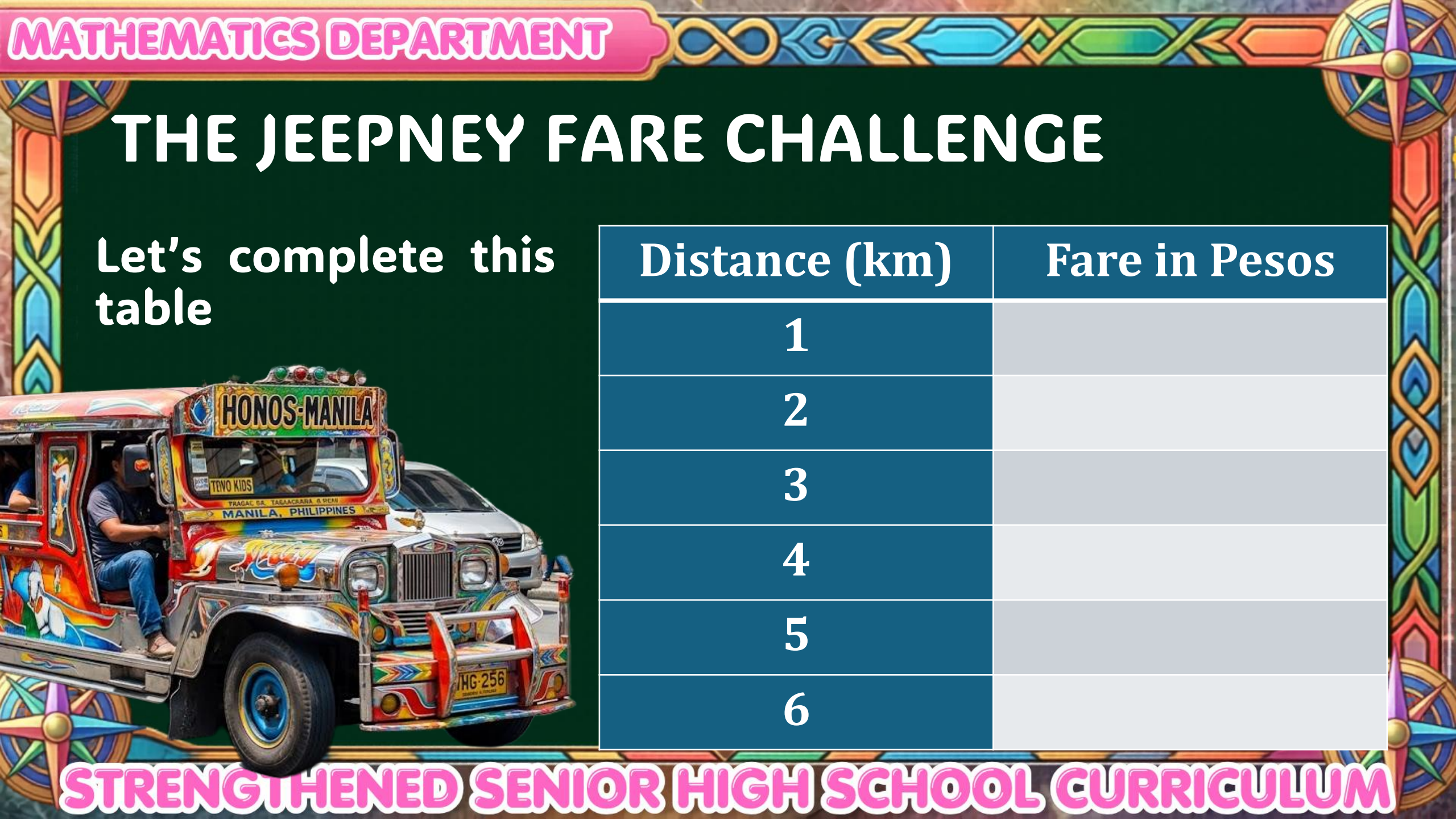
- ① Appreciate the real-world application of piecewise functions by demonstrating persistence and systematic reasoning when breaking down multi-part practical problems.
- ②

THE JEEPNEY FARE CHALLENGE

- Imagine you are riding a traditional jeepney to go to school. The driver charges a flat minimum fare of 13 pesos for the first 5 kilometers.



However, if you travel beyond 5 kilometers, you must pay the 13 pesos minimum plus an additional 1 pesos for every kilometer exceeding the first 5 kilometers.



THE JEEPNEY FARE CHALLENGE

Let's complete this table

Distance (km)	Fare in Pesos
1	
2	
3	
4	
5	
6	



PROCESS QUESTIONS

- What do you notice about the fare for 1-5 kilometers?
- What happens after 5 kilometers?
- Can we write a single, simple linear equation like $y = 2x$ to represent this entire situation for all distances?

Step 1: Define Case 1 (Distances up to 5 km)

Condition: $0 < x \leq 5$

Rule: The fare is always 13 pesos.

Sub-function: $f(x) = 13$

Step 2: Define Case 2 (Distances greater than 5 km)

Condition: $x > 5$

Rule: The base fare of 13 pesos plus 1 peso for every kilometer over 5.

Exceeding distance: $(x - 5)$

Sub-function: $f(x) = 13 + 1(x - 5) = 13 + x - 5 = x + 8$

Step 3: Combine into Piecewise Notation

$$f(x) = \begin{cases} 13, & \text{if } 0 < x \leq 5 \\ x + 8, & \text{if } x > 5 \end{cases}$$

QUESTION:

If you travelled for 12 kilometers, how much will be your fare?

Solution:

$$f(x) = \begin{cases} 13, & \text{if } 0 < x \leq 5 \\ x + 8, & \text{if } x > 5 \end{cases}$$

Since 12 is greater than 5, use $f(x) = x + 8$

$$f(12) = 12 + 8$$

$$f(12) = 20$$

Your total fare is 20 pesos.

**LET'S DO THIS
TOGETHER**



SITUATION 1: Bulk Buy Discount

A local print shop charges ₱5.00 per page for orders up to 50 pages. For orders over 50 pages, the shop offers a discounted rate of ₱3.00 for each additional page beyond 50, while charging a base price of ₱250.00 for the first 50 pages. Write the piecewise function $C(p)$ representing the total cost for p printed pages. If you printed a total of 25 pages, how much will you pay?



**TRY IT ON
YOUR OWN**

SITUATION 2: MALL PARKING FEE

A parking lot in a shopping mall charges ₱40 for the first 2 hours. For every hour (or fraction of an hour) beyond 2 hours, an additional fee of ₱15 per hour is added.

- 1. Write the piecewise function $P(h)$ that represents the total parking fee for h hours.**
- 2. Calculate the total cost if a customer parks for:**
 - a. 1.5 hours**
 - b. 5 hours**

LET'S SUM IT UP!

- What makes a piecewise function different from a standard linear function?
- Why do businesses and transportation systems use piecewise pricing models?



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EXIT TICKET

STRENGTHENED SENIOR HIGH SCHOOL CURRICULUM

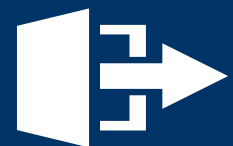
 EXIT TICKET

A local internet provider offers a home fiber plan. The basic plan costs **₱999 per month**, which includes up to **100 GB** of data usage. If a user exceeds 100 GB, an additional fee of **₱10 per extra GB** is charged.

- (5 points)** Write the piecewise function $C(g)$ representing the total monthly bill where g is the data used in GB.



SESSION 2: COMMISSIONS

**EXIT TICKET**

A local internet provider offers a home fiber plan. The basic plan costs **₱999 per month**, which includes up to **100 GB** of data usage. If a user exceeds 100 GB, an additional fee of **₱10 per extra GB** is charged.

- 2. (2.5 points)** Calculate the bill of a household that uses **75 GB** in a month.

**SESSION 2: COMMISSIONS**

 EXIT TICKET

A local internet provider offers a home fiber plan. The basic plan costs **₱999 per month**, which includes up to **100 GB** of data usage. If a user exceeds 100 GB, an additional fee of **₱10 per extra GB** is charged.

3. (2.5 points) Calculate the bill of a household that uses 130 GB in a month.

**SESSION 2: COMMISSIONS**

WAYS FORWARD

Household Rate Investigation:

Analyze one monthly household utility bill (electric, water, or mobile phone plan) and identify if it uses a flat rate or tiered pricing. Have them attempt to draft its piecewise structure in their notebooks.

SESSION DEVELOPER



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